Topology Optimization
Lecture 14

Topology optimization of conductive heat transfer devices: An experimental investigation

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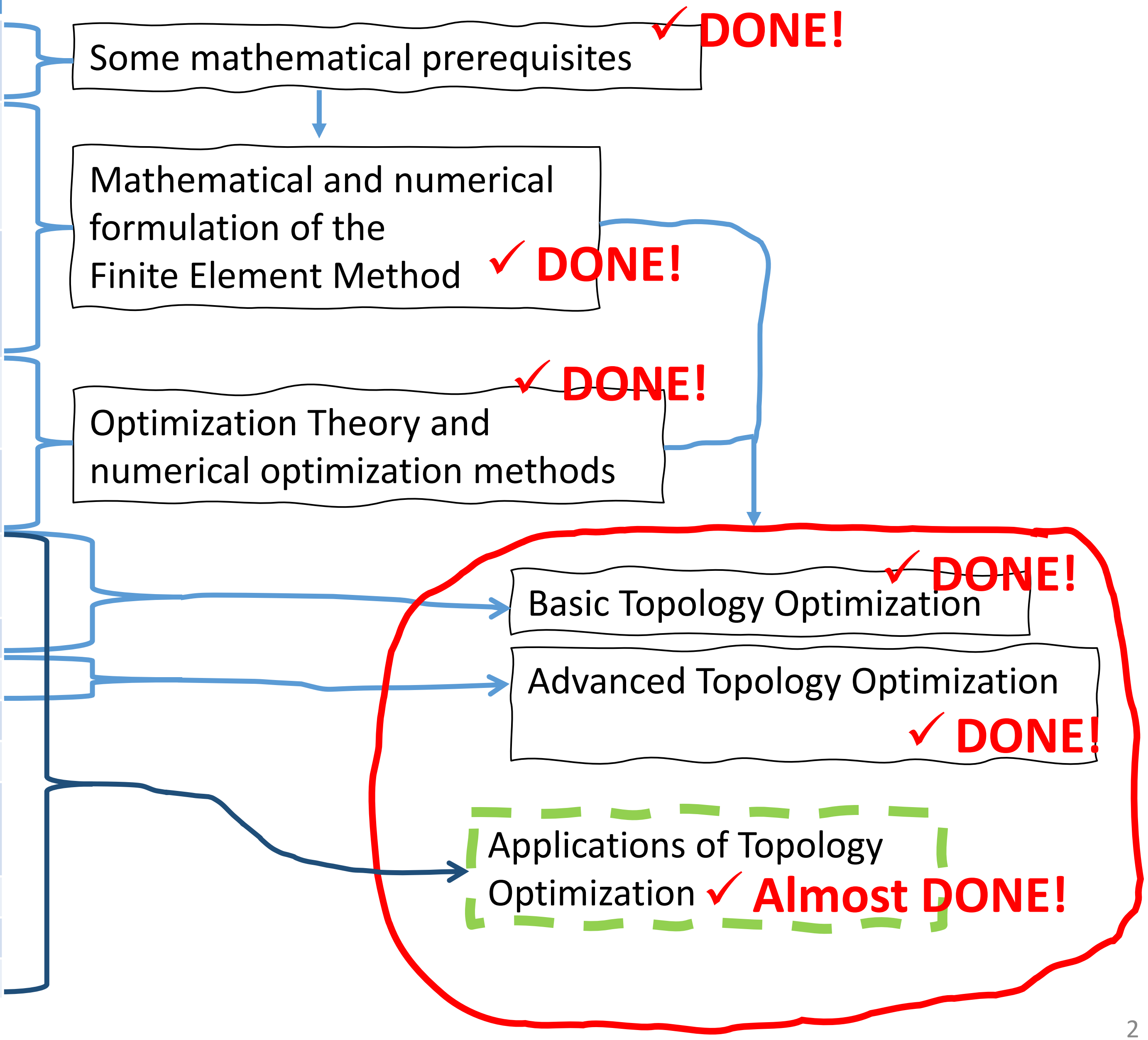
Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



The main points of the paper

- The paper considers a heat conduction problem, where a heat sink is used to lower the temperature of an area being heated by a volumetric heat source.
- The challenge is to place high conductive material in the area to utilize the heat sink to lower the overall temperature.
- Topology Optimization used on this kind of problem result in tree-like structures.
- The paper uses topology optimization to derive two designs with different volume fractions.
- The two designs are manufactured, and measurements are made to investigate the validity of the topology optimized designs.
- CFD simulations are compared to the experimental results.
- The designs were found to be effective in reducing the temperature and the design with a higher volume fraction was found to be more efficient than the low volume fraction design.
  - No big surprises…
  - Should have used robust topology optimization to ease production!

The Topology Optimization Process

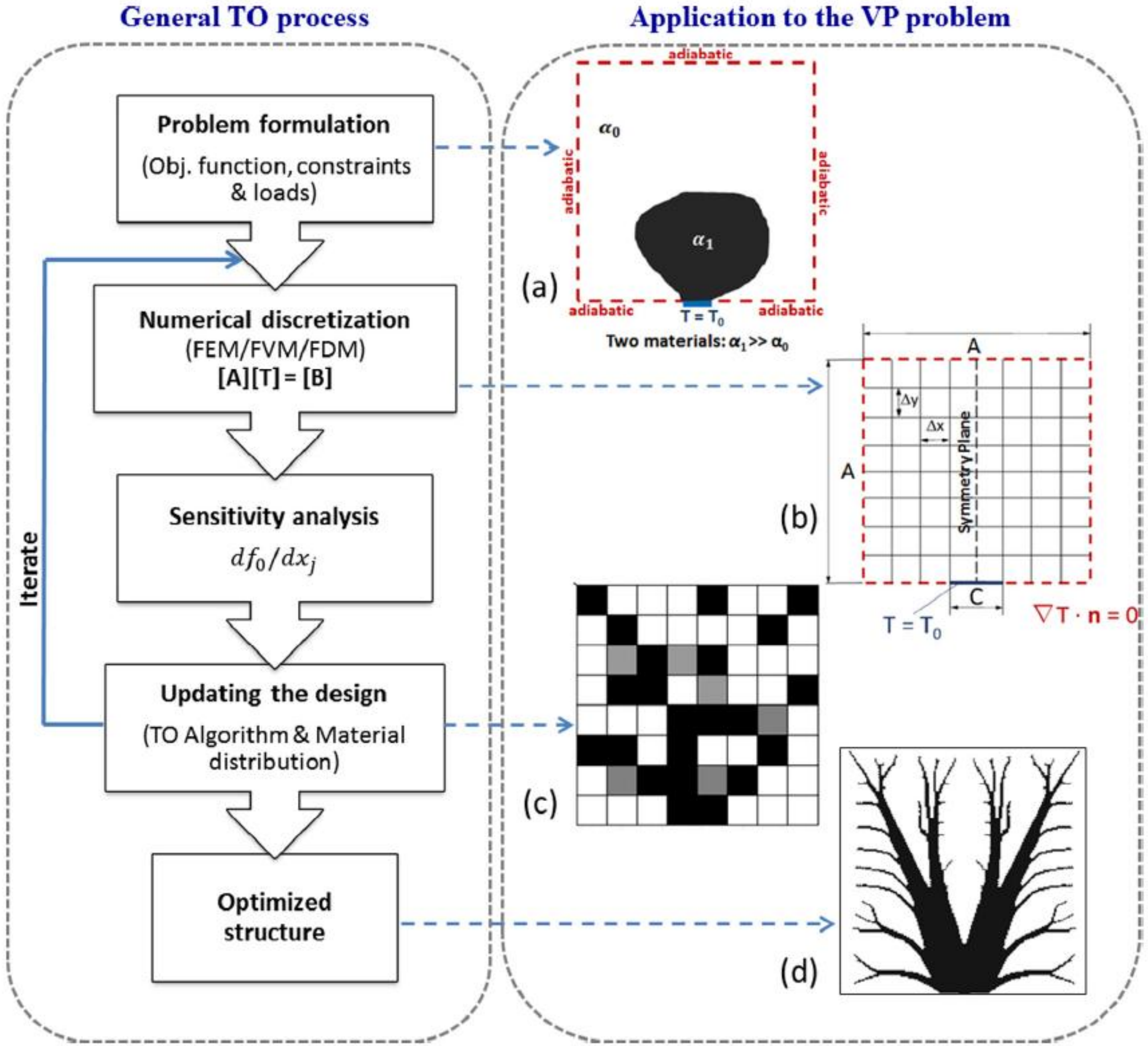


Fig. 1. The topology optimization process.

Minimize/Maximize:

$$f_0(\boldsymbol{x}) \tag{1a}$$

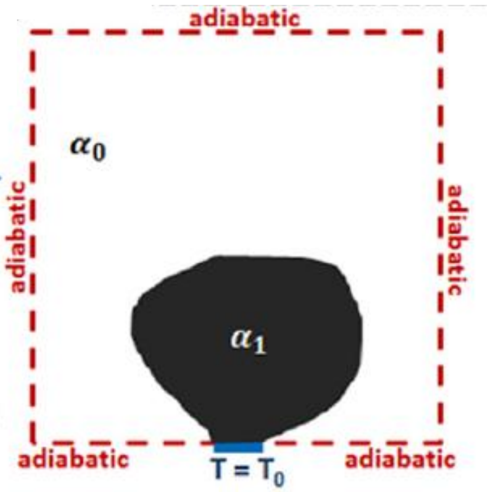
subject to:

$$f_i(\boldsymbol{x}) \leqslant 0, i = 1, 2, \ldots, m \tag{1b}$$
$$R_h(\boldsymbol{x}) = 0, h = 1, 2, \ldots, e \tag{1c}$$
$$\boldsymbol{x} \in \mathbf{X}, \tag{1d}$$

where $\boldsymbol{x} = (x_1, \ldots x_n)^T \in R^n$ are the so called design variables with $\mathbf{X} = \left\{\boldsymbol{x} \in R^n | x_j^{min} \leqslant x_j \leqslant (x_j)^{max}, j = 1, \ldots, n\right\}$ and $f_0, f_1, f_2, \ldots, f_m$ are provided continuously differentiable (at least twice) real valued function on $\mathbf{X}$. x_j^{min} and x_j^{max} belongs to R such that $x_j^{min} \leqslant x_j^{max}\ \forall j$. The design variable vector $\boldsymbol{x}$ is bounded numerically such that its final value represents only one value i.e either x_j^{min} or x_j^{max}. This represents the interpolation of two materials in the design domain which is obtained by a penalization technique.

The constraints $R_1, R_2, \ldots, R_e$ represents the state equations for the physical problem being solved. For example, for a pure steady state heat conduction application, $R_1 = 0$ will be the steady state heat equation. In TO problem applied to fluid flows with no heat transfer $R_1 = 0$ and $R_2 = 0$ will be the mass and momentum conservation equations, respectively.

The Topology Optimization Problem

(a) 

Two materials: $\alpha_1 >> \alpha_0$

- The area (or volume) is heated by a constant volumetric heat source, $\dot{q}$ [W/m^3].
- All boundaries are adiabatic (no heat flux, $\alpha \nabla T \cdot \hat{n} = 0$), except on a small part of the bottom boundary.
- The heat sink is the non-adiabatic part of the boundary and is kept at a constant temperature T_0.
- The heat can only escape from the non-adiabatic part of the boundary.
- The aim is to minimize the average temperature by distribution of a highly conductive material.

Mathematically, the above TO problem can be expressed as follows:

Minimize:

$$f_0 = \frac{1}{M^2}\sum_{j=1}^{M^2} T_j \quad (2a)$$

subject to:

$$\nabla \cdot (\alpha_{eff} \nabla T) + S_{eff}(x, y) = 0 \quad (2b)$$

$$\frac{1}{M^2}\sum_{j=1}^{M^2} x_j \leqslant \phi_{max} \quad (2c)$$

where f_0 is the objective function to be minimized, subjected to a steady state heat Eq. (2b) and an inequality constraint 2c. The

(SIMP) technique [8] is used in the literature. The SIMP technique replaces the discrete integer variables with continuous ones, and then introduces some kind of penalty that forces an evolution of the solution towards discrete 0–1 values as the following:

$$S_{eff} = S_0 + x_j^p (S_1 - S_0) \quad (4a)$$

$$\alpha_{eff} = \alpha_0 + x_j^p (\alpha_1 - \alpha_0) \quad (4b)$$

with $p \geqslant 1$ and the subscript indices 0 and 1 represents two materials of different thermal diffusivities (or conductivities).

$$f_0 = \frac{1}{|\Omega|}\int_\Omega T \, d\Omega$$

T [K] is the temperature field. Ω is the domain and $|\Omega|$ is the domain area or volume,

$$|\Omega| = \int_\Omega 1 \, d\Omega .$$

α [m^2/s] is thermal diffusivity, given by

$$\alpha = \frac{k}{\rho C_p}$$

where k [W/m/K] is thermal conductivity, ρ [kg/m^3] is density, C_p [J/kg/K] is specific heat capacity.

S [WK/J] is thermal source, given, by

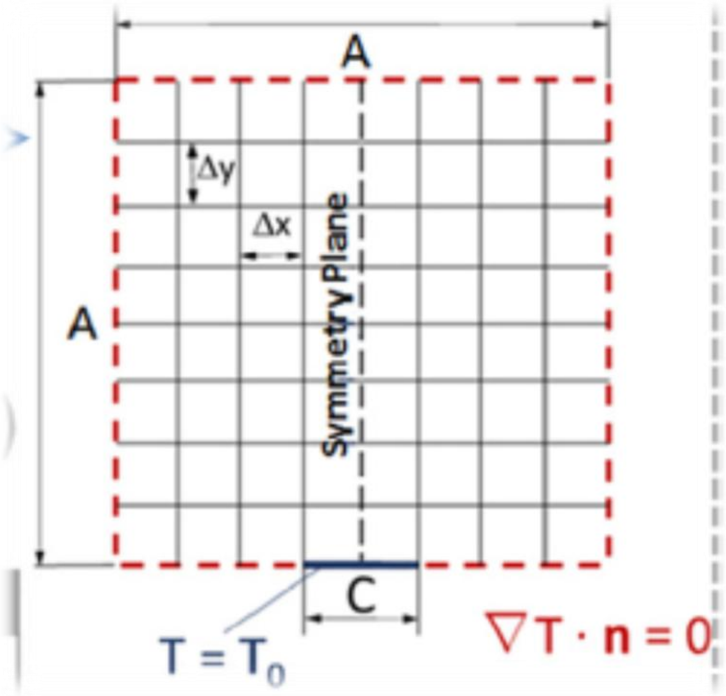
$$S = \frac{\dot{q}}{\rho C_p}$$

$$\frac{1}{|\Omega|}\int_\Omega x \, d\Omega \leq \phi_{max}$$

x is the design variable field.

The Topology Optimization Problem

- The optimization problem is solved using MMA with gradients calculated by the adjoint method.
- The used geometry is a square with side length A=0.1 m. The thickness e<<A is negligible and a 2D model is used.
- The discretization uses $M^2 = 140^2$ square elements with $\Delta x = \Delta y = \frac{100\,mm}{140} = \frac{5}{7}\text{mm} = 0.714\text{ mm}.$
- The heat sink size is C=0.05A=5 mm.
- They state that the densities are filtered, but this is not shown in the problem statement.
  - The filter radius is 1.428 mm, 2x the mesh size.
- They do not mention the use of projection.
- The thermal diffusivities has the ratio $\gamma = \frac{\alpha_1}{\alpha_0} = 760.$
- The SIMP exponent *p* is updated during the optimization process:
  - $S_{eff} = S_0 + x^p(S_1 - S_0)$, $\alpha_{eff} = \alpha_0 + x^p(\alpha_1 - \alpha_0)$
  - They likely start at *p*=1.
  - *p* is ramped up to *p*=4 over 40 MMA iterations.
  - Afterwards, 30 MMA iterations are used at *p*=4.
- The steady state heat equation is solved using OpenFOAM with a finite volume discretization.



Results

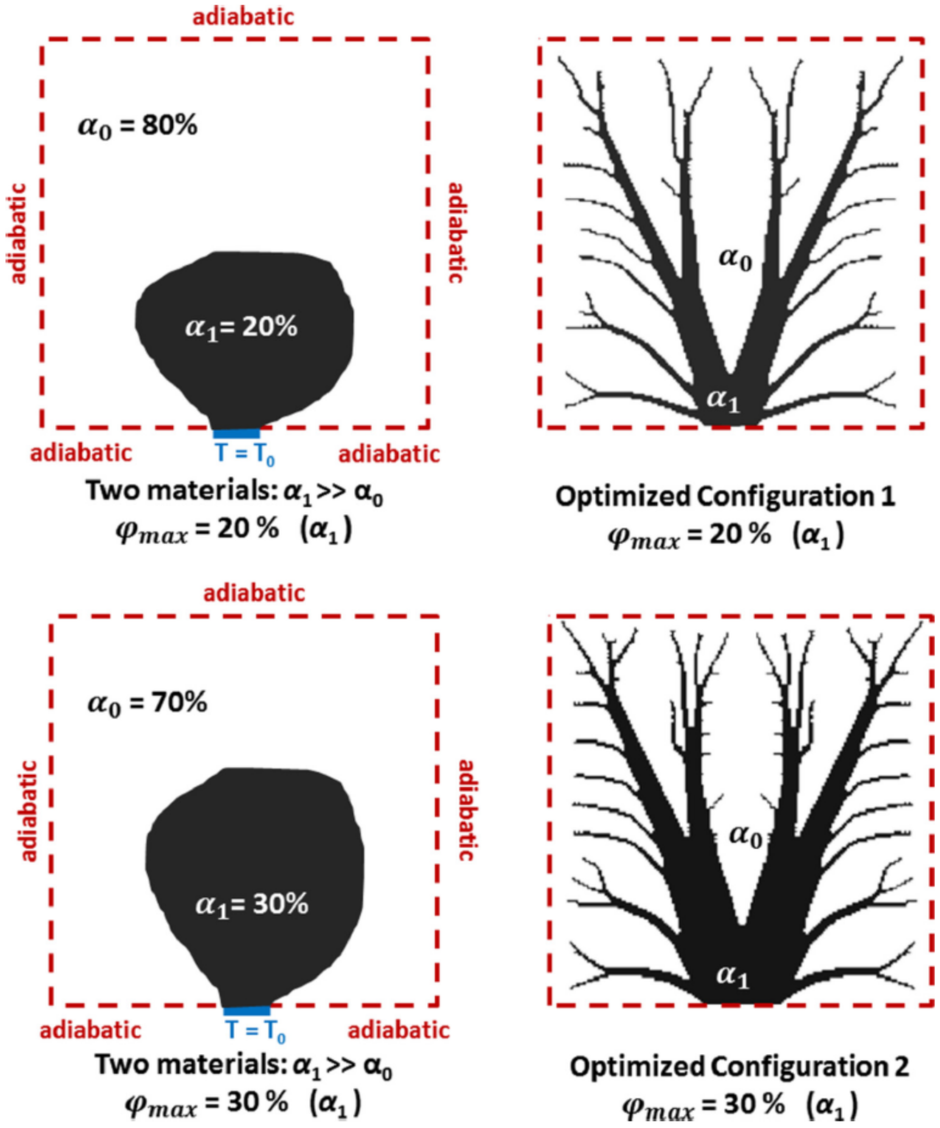


Fig. 2. 2D heat conduction topology optimization: two configurations of the VP problem – $\phi_{max} = 20\%$ and 30%.

Table 1
Material properties for aluminum and epoxy resin polymer.

Material properties	Aluminium	Epoxy resin polymer
Thermal conductivity, k (W m^{-1} K^{-1})	214.04	0.19663
Density, ρ (kg m^{-3})	2700	1100
Specific heat, C_p (J kg^{-1} K^{-1})	891	1522.72
Thermal diffusivity, α (m^2 s^{-1})	8.89×10^{-5}	1.17×10^{-7}

Units missing!!

$$\gamma = \frac{\alpha_1}{\alpha_0} = \frac{8.89 \times 10^{-5}}{1.17 \times 10^{-7}} = 759.82 \simeq 760 \quad (5)$$

$$\dot{q} = \frac{\dot{q}_{\text{surface}}}{\text{thickness}} = \frac{588\ \text{W/m}^2}{e} = \text{“some number” W/m}^3$$

$\dot{q}$ is the input to the steady state heat equation

$$\nabla \cdot (\alpha \nabla T) + S = 0 \Rightarrow -\nabla \cdot (\alpha \nabla T) = S = \frac{\dot{q}}{\rho C_p}$$

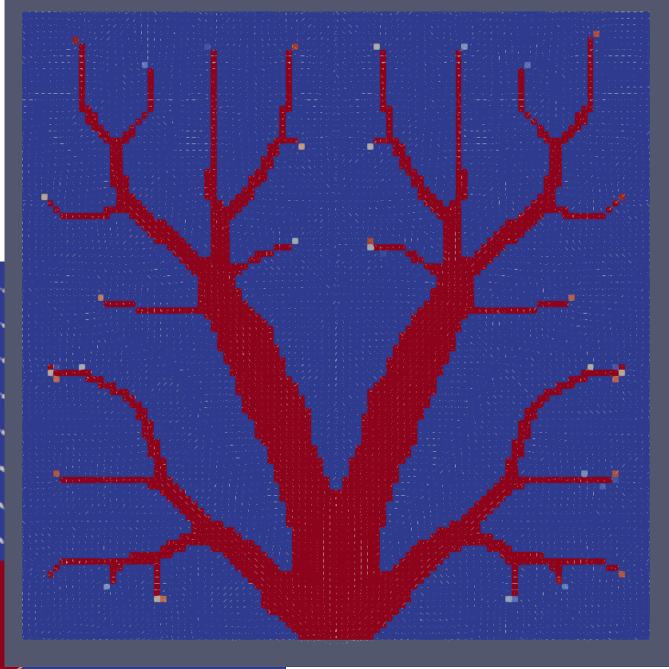
$$S = S_0 + x^p(S_1 - S_0) = \frac{\dot{q}}{\rho_0 C_{p0}} + x^p\left(\frac{\dot{q}}{\rho_1 C_{p1}} - \frac{\dot{q}}{\rho_0 C_{p0}}\right)$$

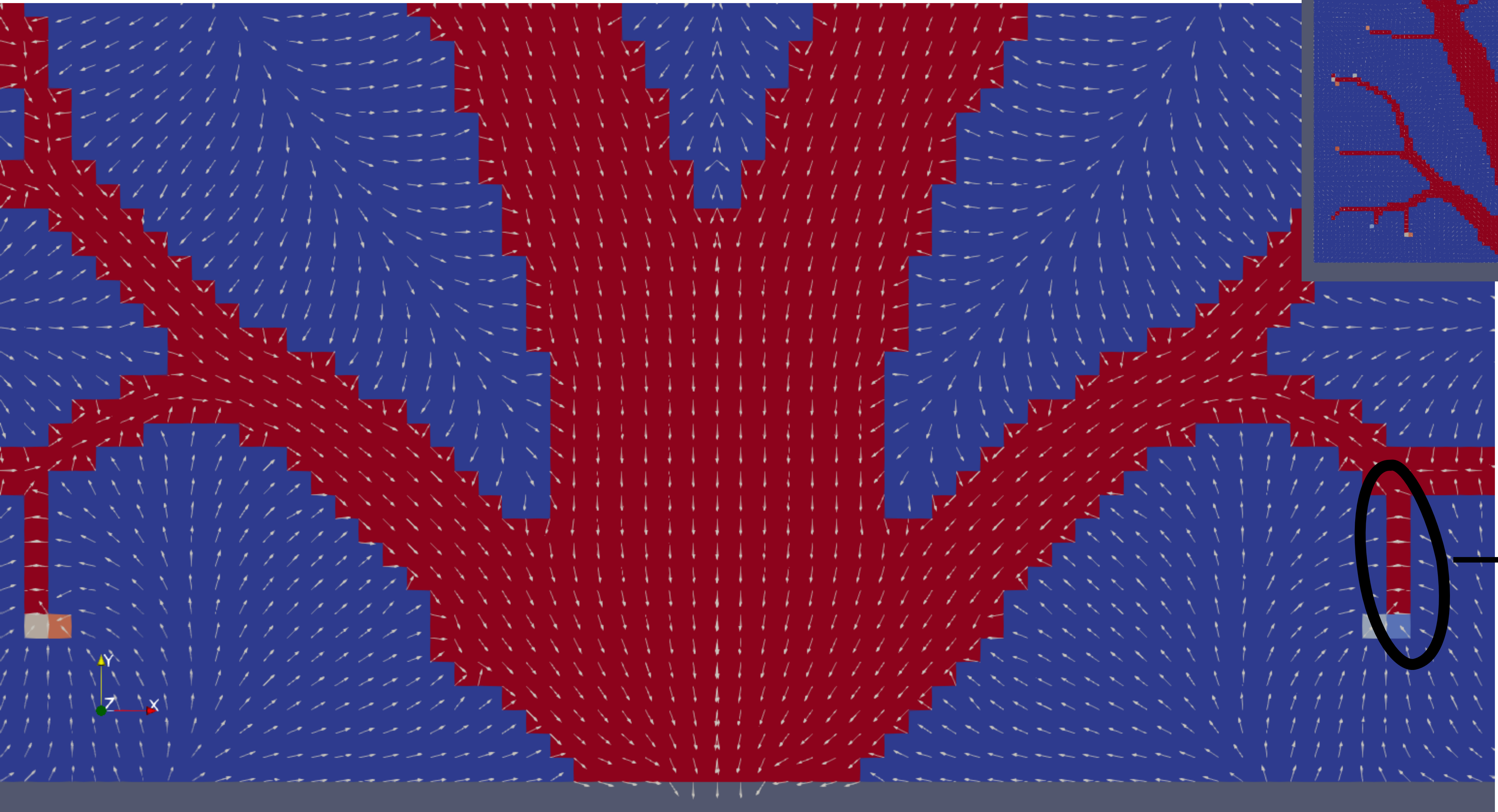
Changing $\dot{q}$ scales the input and hence scales the entire solution temperature.

With the cost function being the average temperature, this corresponds to a rescaling of the cost function, which does not change the solution of the optimization problem.

In a non-linear optimization problem with multiple local optima, the scaling combined with the numerical tolerances used in the optimization method may, however, in practice result in convergence to a different optimum.

Arrows show the heat flux direction given by Fourier's Law: $\vec{q} = -k\nabla T \Rightarrow \vec{q} \propto -\nabla T/|\nabla T|$





Thin feature ☹
$\Downarrow$
One should use Robust Topology Optimization ☺

Previous:

On projection methods, convergence and robust formulations in topology optimization
F. Wang, B. S. Lazarov & O. Sigmund
Struct Multidisc Optim 43, 767–784 (2011)
https://doi.org/10.1007/s00158-010-0602-y

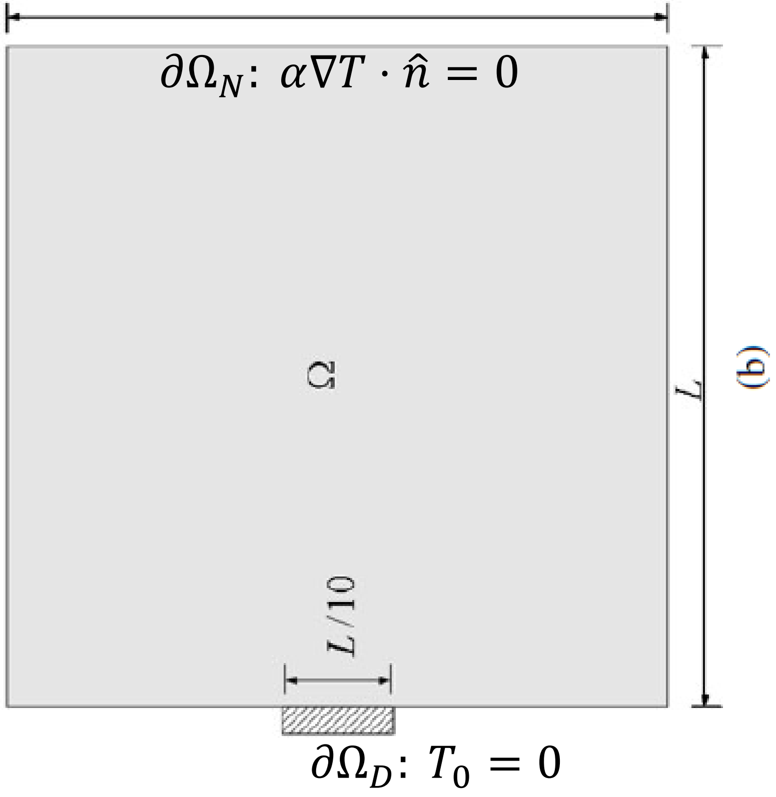
- A heat conduction problem was also considered:

$$
\begin{aligned}
\min_{\rho} &: f(\rho) = \mathbf{l}^{\mathrm{T}}\mathbf{u} \\
s.t. &: \mathbf{K}\mathbf{u} = \mathbf{f} \\
&: f_v(\rho) = \frac{\sum_{i \in \mathbb{N}_e} \bar{\tilde{\rho}}_i v_i}{V} \leq V^* \\
&: \mathbf{0} \leq \rho_i \leq \mathbf{1} \forall i \in \mathbb{N}_e
\end{aligned}
\qquad (1)
$$

with **l**=**f** where **f** is the load vector and **u** is the discretized temperature degrees of freedom, with the relation

$$T(x, y) = \sum_i u_i \phi_i(x, y) = u_i \phi_i(x, y)$$

with basis functions $\phi_i(x, y)$. Zero Dirichlet, $T_0 = 0$ at Ω_D and adiabatic Neumann b.c.'s were used.



$$\nabla \cdot (\alpha_{\mathit{eff}} \nabla T) + S_{\mathit{eff}}(x, y) = 0$$

- This give the cost function

$$f(\rho) = \mathbf{f}^T \boldsymbol{u} = \sum_i \mathrm{f}_i u_i = \mathrm{f}_i u_i$$

with f_i being the load vector components, i.e.

$$\mathrm{f}_i = \int S_{eff}(x, y)\phi_i(x, y) d\Omega \quad \text{and}$$

$$f(\rho) = \mathrm{f}_i u_i = \int S_{eff}(x, y)\phi_i(x, y) u_i d\Omega = \int S_{eff}(x, y) T(x, y) d\Omega \quad \text{using } T(x, y) = u_i \phi_i(x, y).$$

- The integral $\int S_{eff}(x, y) T(x, y) d\Omega$ is named the thermal compliance.
- The nice thing about minimizing the thermal compliance is that the adjoint variable, λ, is found by solving the same problem as the forward problem, but with zero Dirichlet conditions, which is not always assumed in the forward problem. This result in the solution $\lambda = T - T_0$ and no additional FEM problem needs to be solved to find the gradient of the thermal compliance cost function.

- If *S* is a constant, the thermal compliance is just a scaling of the average temperature.

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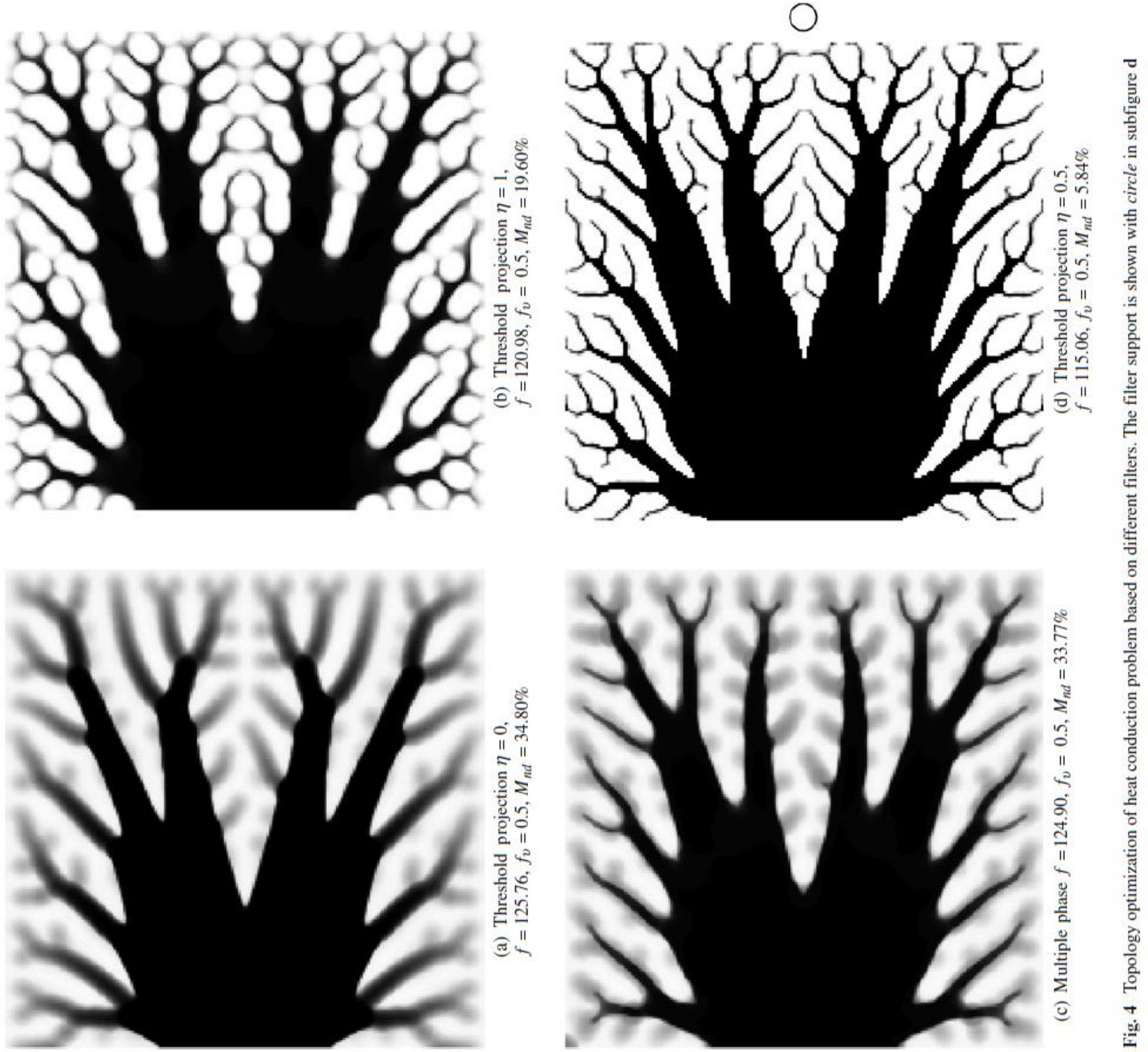
- Tree structures are found.
- The volume fraction used here is 0.5, whereas today's paper uses 0.2 and 0.3.
- The exact thicknesses of the branches are not well controlled with the standard filtering and projection.
- Robust topology optimization enables much better control over the geometry length scales/feature sizes, as seen on the next slide.



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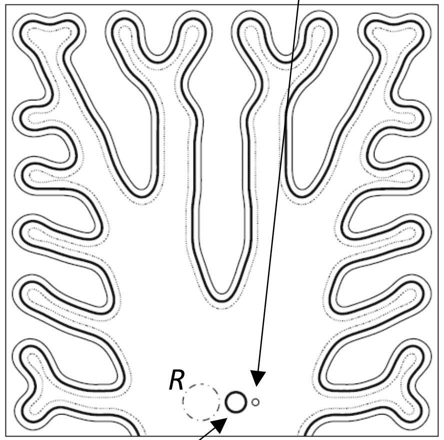
η is used as
$\eta_d = \eta,\ \eta_i = 0.5,\ \eta_e = 1 - \eta$
and determines the difference between the 3 designs and thus the “manufacturing tolerance”:



Feature size, b, is determined both by filter radius R and η:

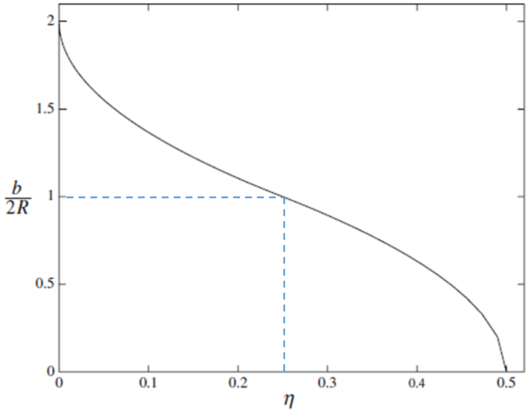


Fig. 12 Normalized length scale on the intermediate design as a function of the threshold η for the robust formulation

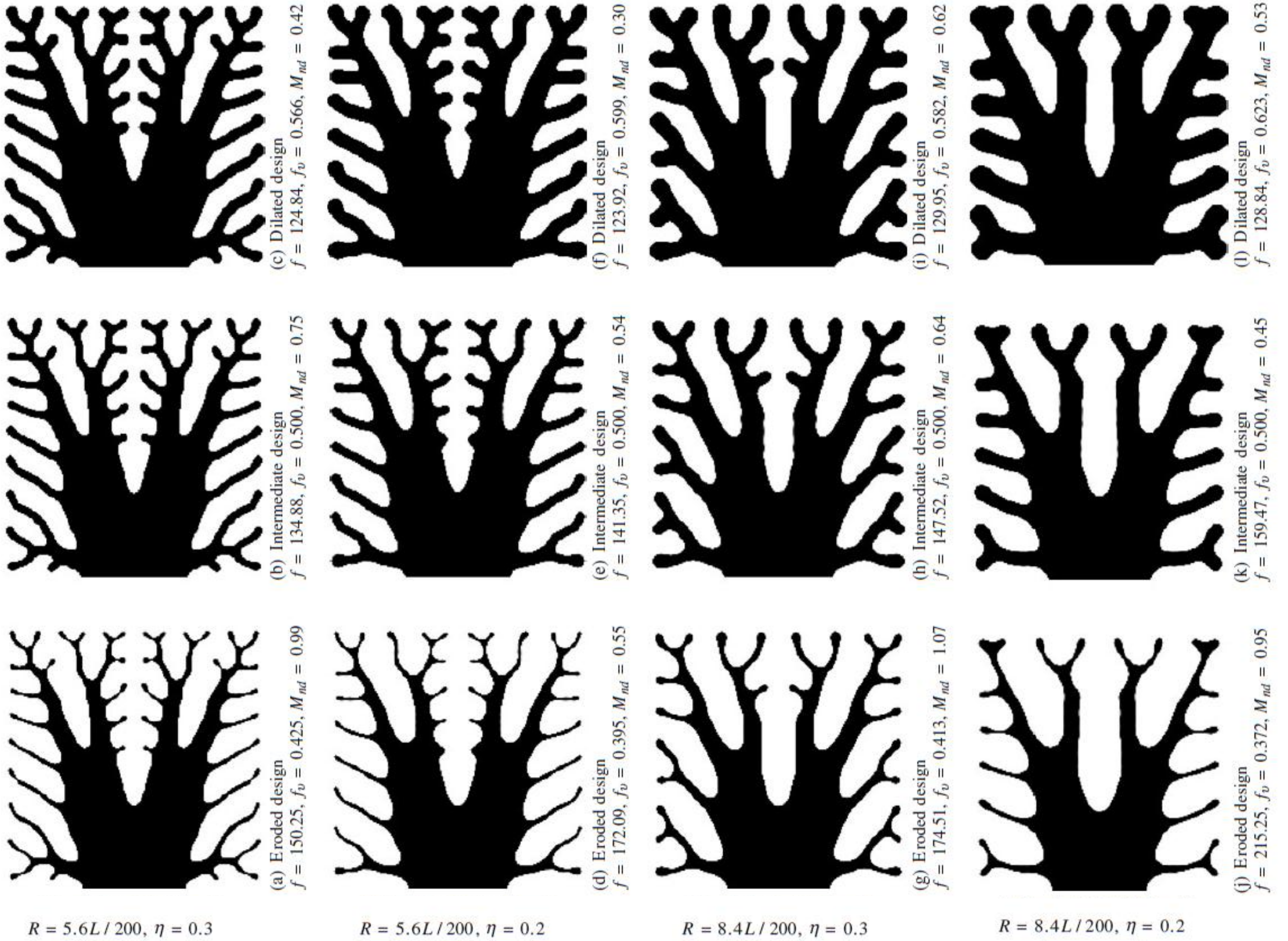
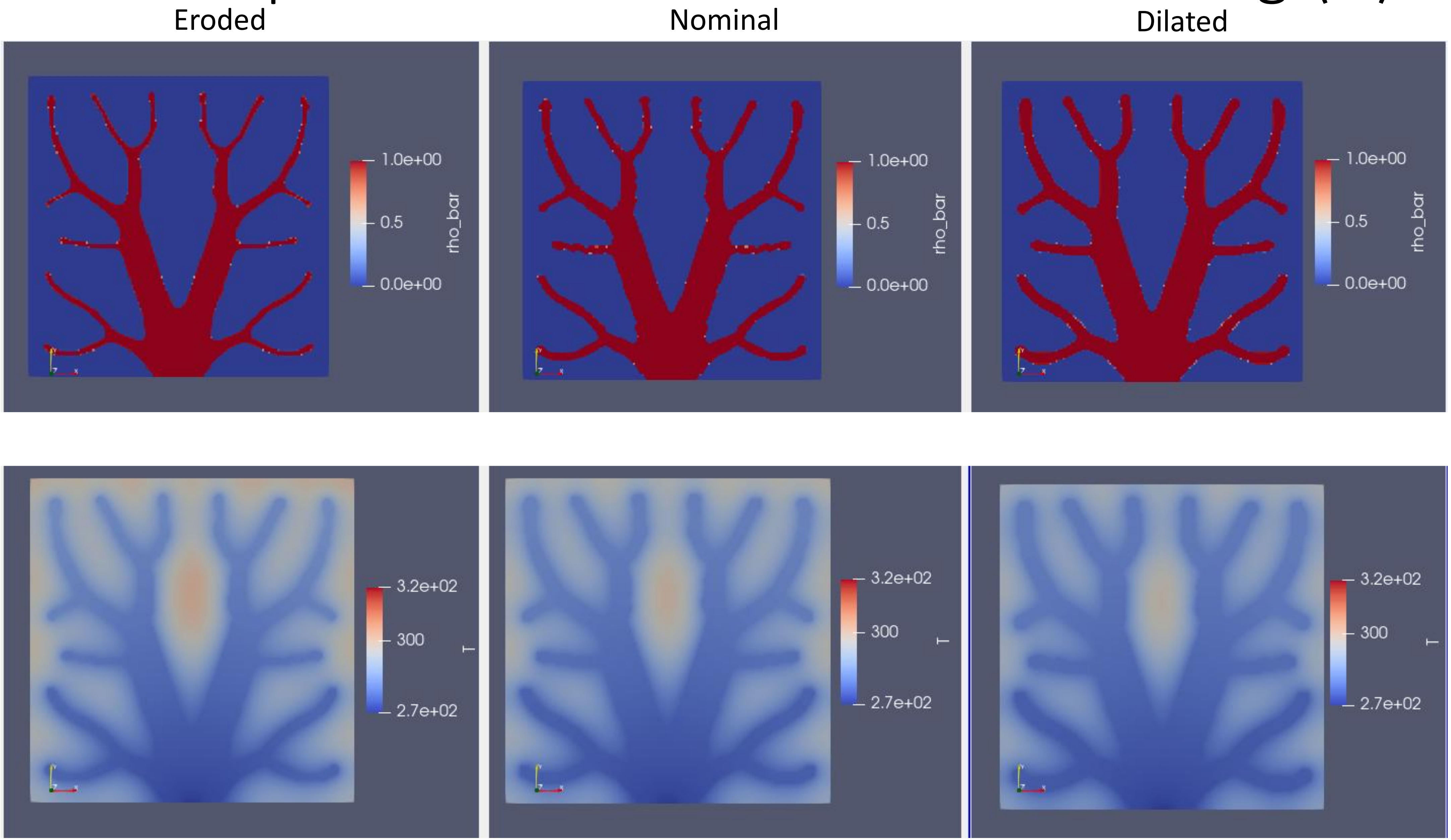


Fig. 7 Robust heat conduction design for different parameter sets

Robust optimization in FEniCS for minimizing $\langle T \rangle$



Fabrication

- They mention that grey areas (densities not being 0 or 1) always exist in the optimized design.
- The optimized design then needs to be post-processed to remove the intermediate densities.
- The post-processing change the volume fractions from 0.2 and 0.3 to 0.179 and 0.286, respectively.
- This is a sign that they did not use projection after filtering directly in the optimization process, since this removes most of the grey areas.
- Laser cutting and water jet cutting was used, and the water jet cutting was found to be superior.
- With all the small structures, a robust optimization might have made the fabrication easier and more manufacturing tolerant.
- A low conductive polymer was molded around the much more conductive aluminum tree structure.

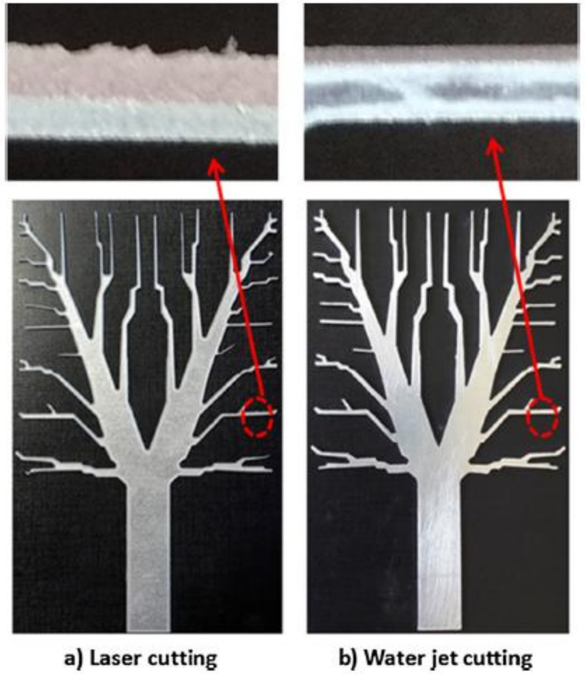


Fig. 4. Tree structures obtained from: (a) Laser cutting and (b) Water-jet cutting.

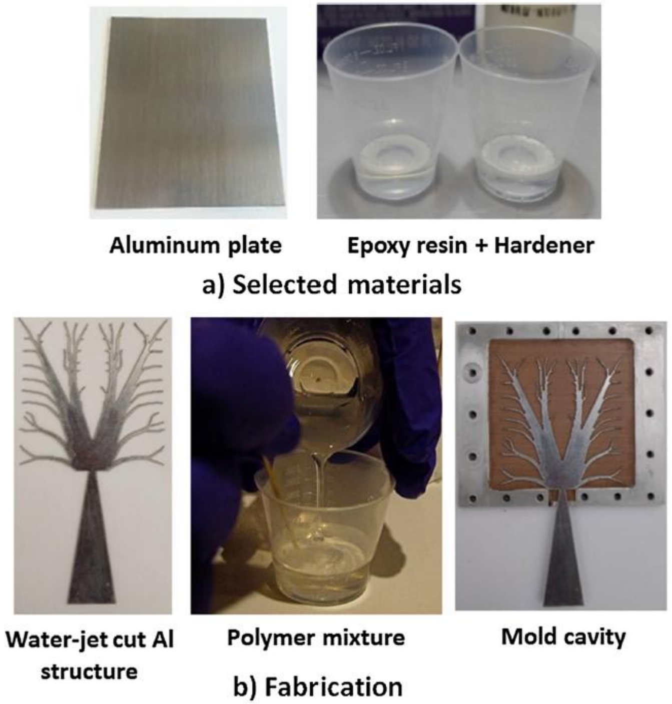


Fig. 5. Fabrication process.

Fabrication

- The aluminum tree with surrounding polymer.
- One side of the aluminum is fitted with a heating device, to approximate the volumetric heating used in the topology optimization.
- The other side is painted with a high emissivity coating to enhance IR temperature measurements.
- The bottom triangle is kept a 0 °C to "implement" the Dirichlet boundary conditions used in the optimization and simulation.

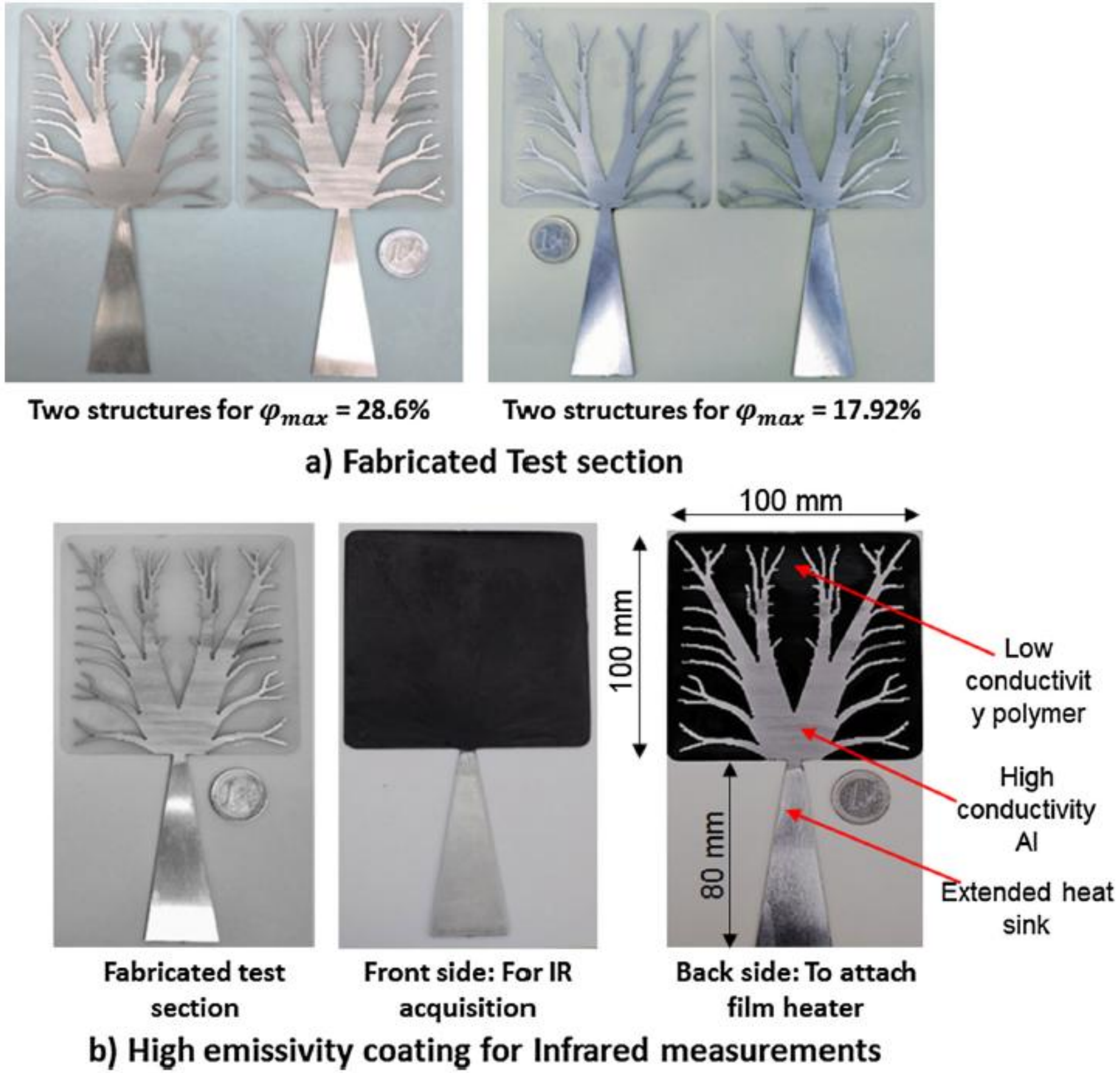


Fig. 6. Fabricated test section.

Experimental setup

- Measurements were done at steady state, after ~15 hours.

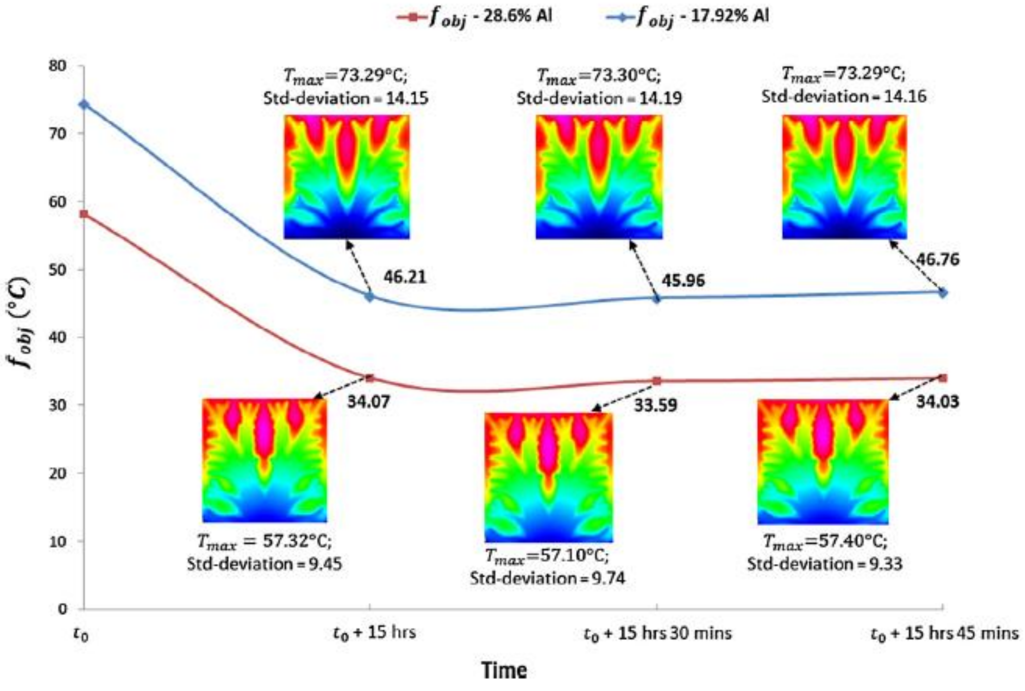


Fig. 9. IR images acquired at steady state for $\phi_{max} = 28.6\%$ and $\phi_{max} = 17.92\%$ with $T_0 = 0\,°C$.

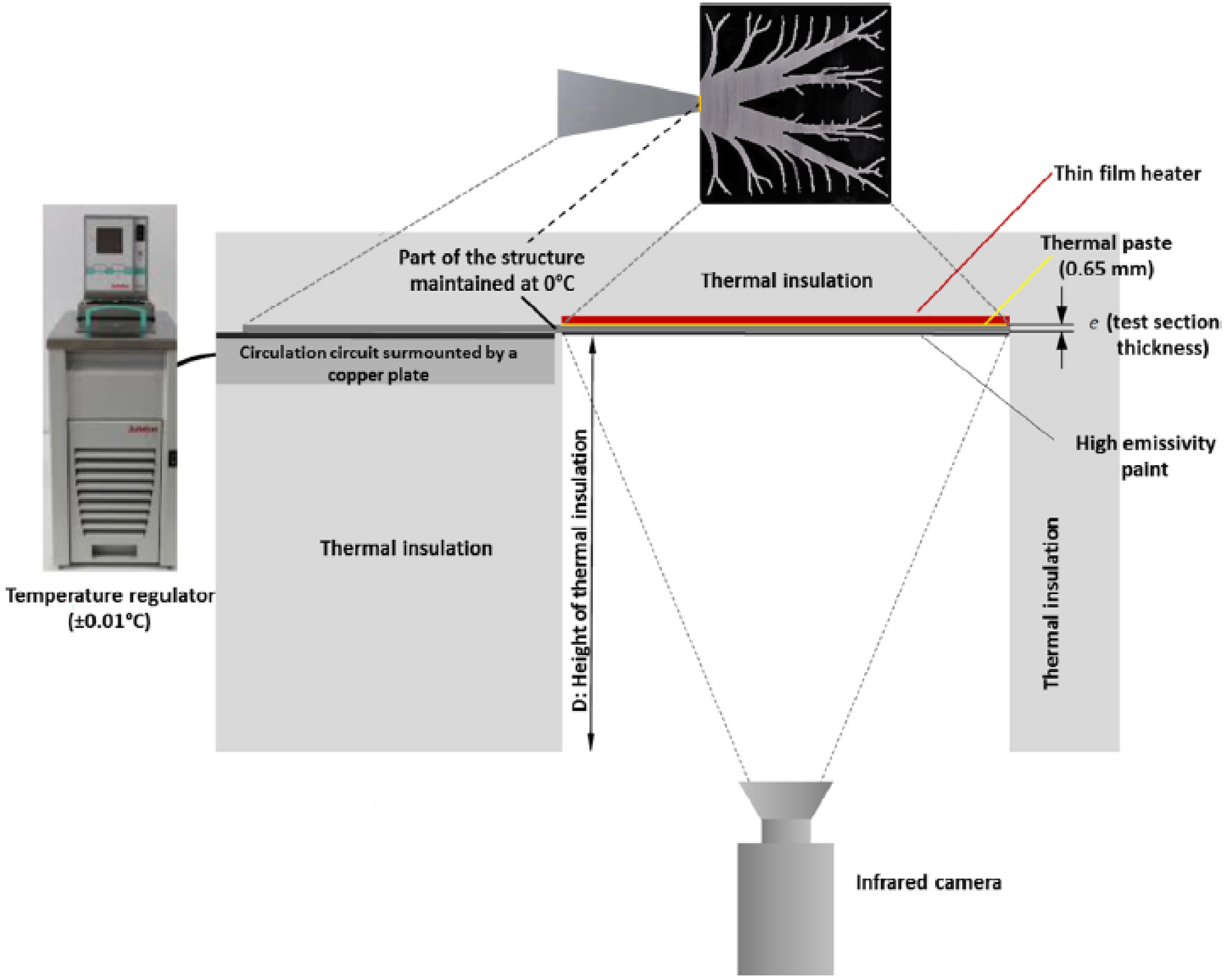


Fig. 7. Schematic representation of the experimental test bench.

CFD Simulations

- 3D CFD simulations of the system were performed.
- This is a more accurate model than the one used in the topology optimization.

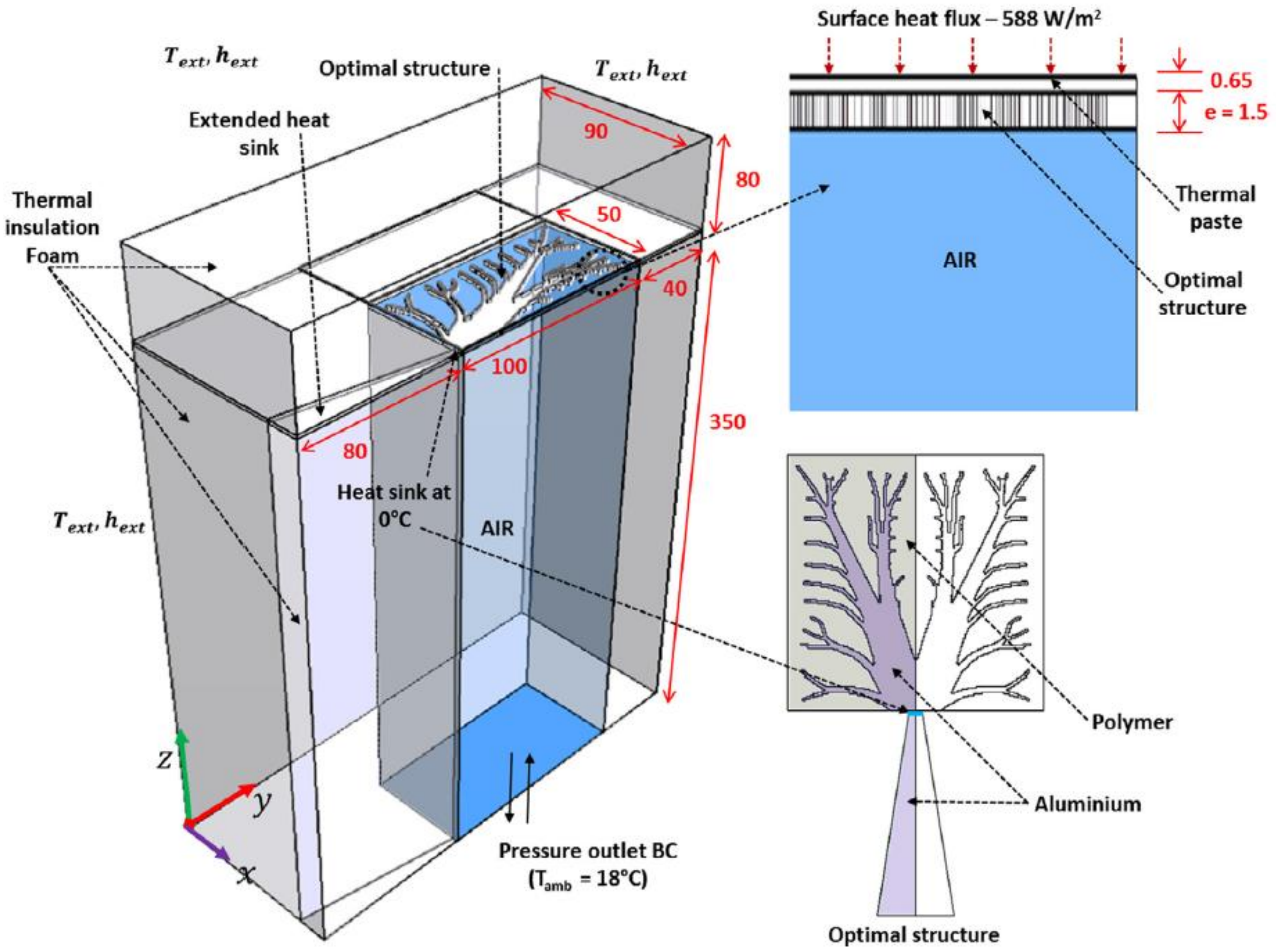


Fig. 10. Axisymmetric computational domain in mm.

CFD Simulations

- The CFD simulations compares reasonably well with the measurements.
- Using the CFD simulations directly in the optimization might have led to a different design.
- Since the CFD model is more realistic than the simple conductive model, a design optimized using the realistic model should potentially perform better in reality.

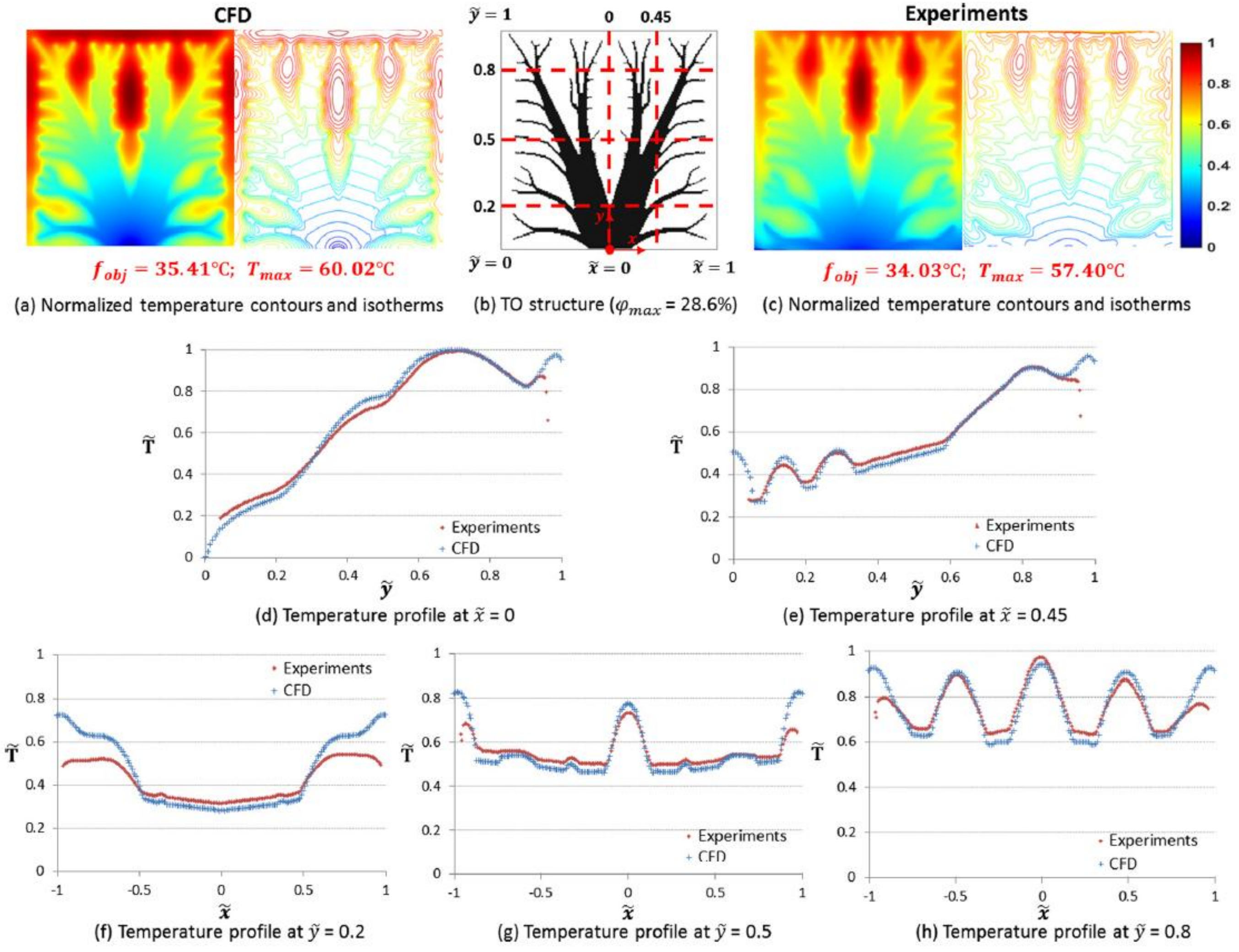
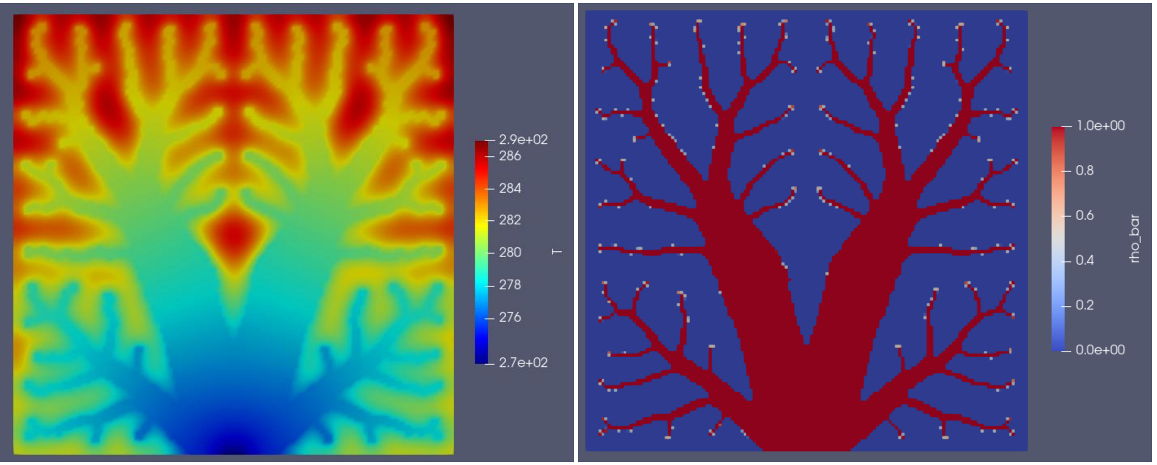


Fig. 12. Normalized steady state temperature measurement for the topology optimized structure with $\phi_{max} = 28.6\%$ and $T_0 = 0$ °C.

Homebrewed non-robust topology optimized design



CFD Simulations

- The ~30% volume fraction design is performing better in the experiments, as it should.

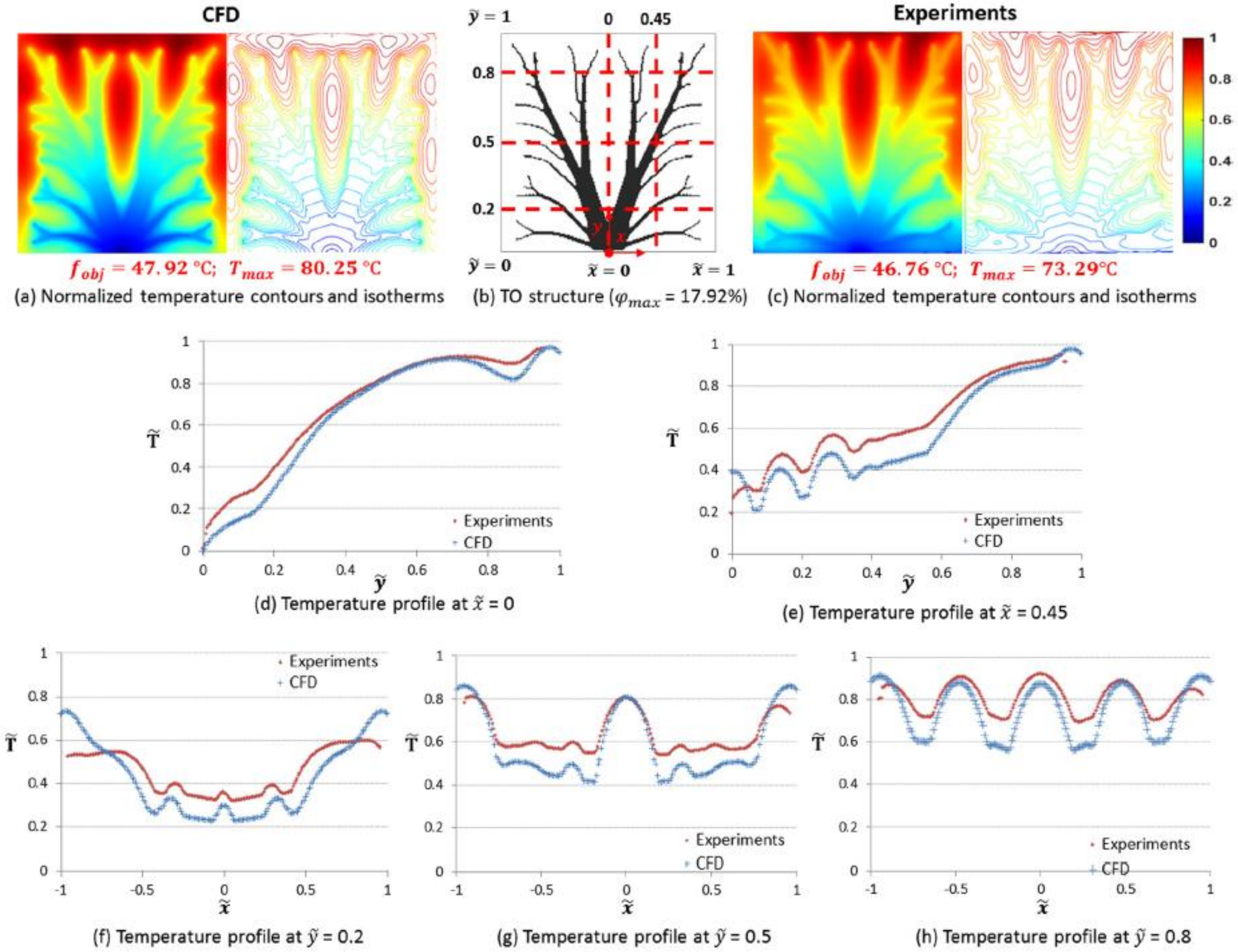


Fig. 16. Normalized steady state temperature measurement for the topology optimized structure with ϕ_{max} = 17.92% and T_0 = 0 °C.

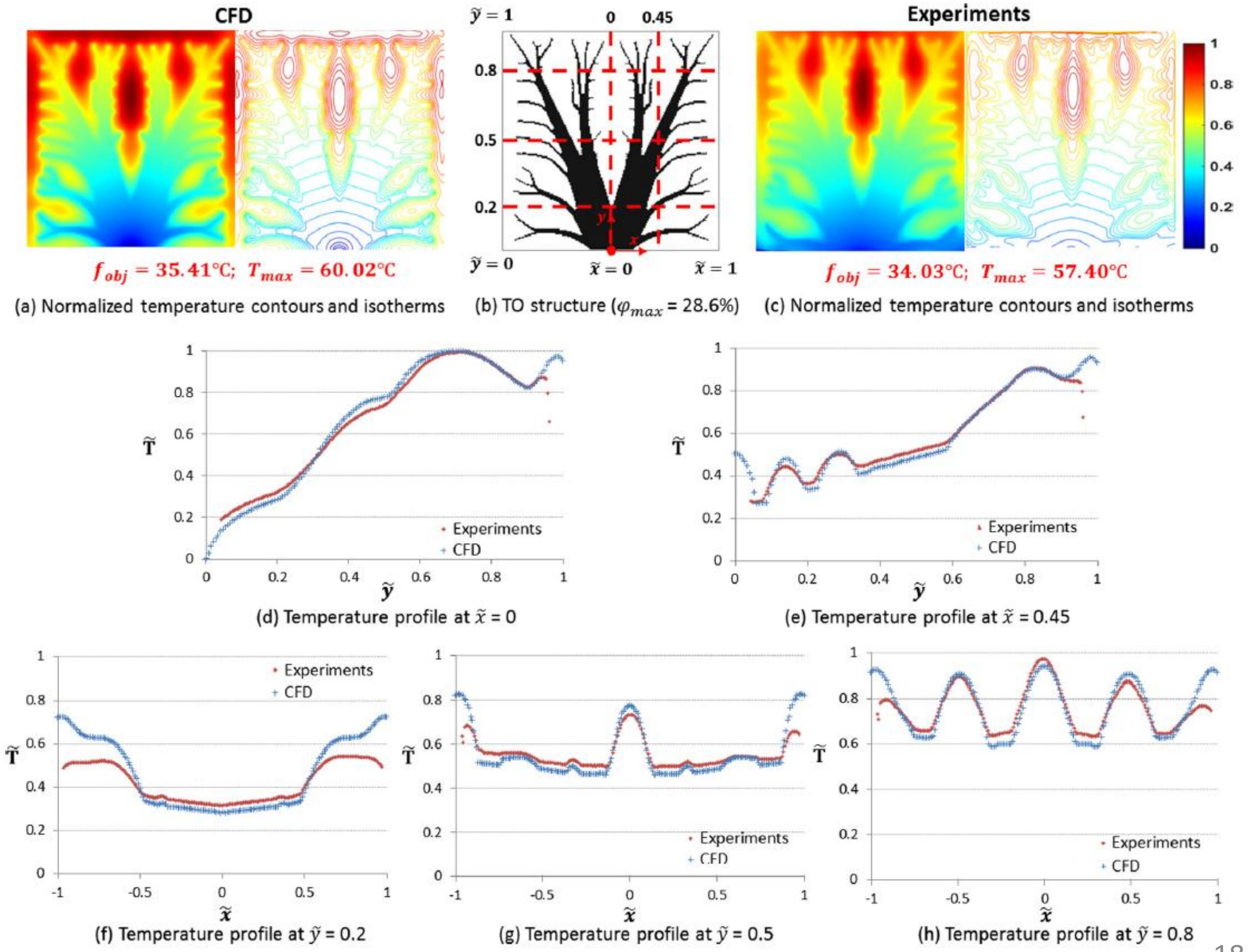


Fig. 12. Normalized steady state temperature measurement for the topology optimized structure with ϕ_{max} = 28.6% and T_0 = 0 °C.

Conclusions

- A heat conducting device was designed using Topology optimization.
- A simple filter scheme was used, but not any projection to create a 0-1 design. Post-processing was used instead to remove intermediate densities.
- Two designs were fabricated and experimentally characterized. As expected, the high-volume fraction design performed the best.
- CFD simulations were performed to compare with measurements and an overall good agreement was found.
- The designs were found to be effective in reducing the temperature in the given area.
- It could have been interesting to see:
  - A robust optimized design with easier manufacturing, e.g., yielding volume fractions of produced design matching those used in the optimization.
  - Comparison of the topology optimization objective values with the CFD objective values. Did the model used in the optimization resemble reality?
  - Topology optimization using a more extensive model considering, e.g., the natural convection in the system.
  - 3D topology optimization combined with 3D metal printing.